

TEMCO Bank Finance & Accounting System

Complete Database Architecture Document v2.0

SECTION 1: YOUR CORE TABLES

1.1 Chart of Accounts (Recursive - Your Design)

chart_of_account	
id	BIGINT PK AUTO_INCREMENT
code	VARCHAR(50) UNIQUE NOT NULL
account_name	VARCHAR(255) NOT NULL
parent_id	BIGINT FK → chart_of_account.id (SELF-REFERENCE)
account_nature	ENUM('DEBIT', 'CREDIT') NOT NULL
is_posting	BOOLEAN DEFAULT TRUE
is_active	BOOLEAN DEFAULT TRUE
created_at	DATETIME
updated_at	DATETIME
created_by	BIGINT FK → user_login.id
updated_by	BIGINT FK → user_login.id

INDEXES:

- PRIMARY KEY (id)
- UNIQUE (code)
- INDEX idx_coa_parent (parent_id)
- INDEX idx_coa_active (is_active)

CONSTRAINTS:

- FK parent_id → chart_of_account.id ON DELETE RESTRICT
- parent_id = NULL for root accounts (5 roots only)

Root Accounts (Seed Data)

ID	Code	Account Name	Parent	Nature	is_posting
1	1000	ASSETS	NULL	DEBIT	FALSE
2	2000	LIABILITIES	NULL	CREDIT	FALSE
3	3000	EQUITY	NULL	CREDIT	FALSE
4	4000	REVENUE	NULL	CREDIT	FALSE
5	5000	EXPENSES	NULL	DEBIT	FALSE

Hierarchy Example

```
ASSETS (1000) parent_id=NULL, is_posting=FALSE
├─ Current Assets (1100) parent_id=1, is_posting=FALSE
│   ├── Cash & Bank (1110) parent_id=1100, is_posting=FALSE
│   │   ├── Petty Cash (1111) parent_id=1110, is_posting=FALSE
│   │   │   └─ Head Office Petty Cash (1111-01) parent_id=1111, is_posting=TRUE ← POSTING
│   │   └─ Bank Accounts (1112) parent_id=1110, is_posting=FALSE
│   │       ├── BOC Current A/C (1112-01) parent_id=1112, is_posting=TRUE ← POSTING
│   │       └─ HNB Savings A/C (1112-02) parent_id=1112, is_posting=TRUE ← POSTING
│   └─ Accounts Receivable (1120) parent_id=1100, is_posting=FALSE
│       ├── Trade Receivables (1121) parent_id=1120, is_posting=TRUE ← POSTING
│       └─ Other Receivables (1122) parent_id=1120, is_posting=TRUE ← POSTING
└─ Non-Current Assets (1200) parent_id=1, is_posting=FALSE
    └─ Fixed Assets (1210) parent_id=1200, is_posting=FALSE
        ├── Land & Buildings (1211) parent_id=1210, is_posting=TRUE ← POSTING
        └─ Vehicles (1212) parent_id=1210, is_posting=TRUE ← POSTING
```

...UNLIMITED DEPTH...

1.2 Credit Or Debit (Lookup Table)

credit_or_debit

id	INT PK AUTO_INCREMENT
type	VARCHAR(10) UNIQUE NOT NULL
is_active	BOOLEAN DEFAULT TRUE

SEED DATA:

ID	Type
----	-----
1	DEBIT
2	CREDIT

1.3 Voucher Model (Your Design - Complete)

1.3.1 Voucher (Main Document)

voucher	
id	INT PK AUTO_INCREMENT
voucher_number	VARCHAR(50) UNIQUE
amount	DOUBLE
date	DATETIME
description	TEXT
voucher_type_id	INT FK → voucher_type.id NOT NULL
voucher_status_id	INT FK → voucher_status.id NOT NULL
bank_account_id	INT FK → bank_account.id
general_user_profile_id	INT FK → general_user_profile.id
org_departments_id	INT FK → org_departments.id
spot_member_id	INT FK → spot_member.id
login_session_id	INT FK → login_session.id
is_active	TINYINT DEFAULT 1
updated_at	DATETIME

INDEXES:

- PRIMARY KEY (id)
- UNIQUE (voucher_number)
- INDEX idx_voucher_type (voucher_type_id)
- INDEX idx_voucher_status (voucher_status_id)

- INDEX idx_voucher_date (date)

1.3.2 Voucher Type (Extensible via Data)

voucher_type	
id	INT PK AUTO_INCREMENT
name	VARCHAR(100) NOT NULL
code	VARCHAR(20) UNIQUE
description	TEXT
is_active	BOOLEAN DEFAULT TRUE

EXISTING DATA:

ID	Name	Code
1	Supplier Payment	SP
2	Loan Request	LR
3	Over Draft	OD
4	Share Withdrawal	SW
5	Profit Withdrawal	PW
6	Deposit Withdrawal	DW
7	Voucher installment payment	VIP
8	Revenue payment	RP
9	Penalty payment	PP
10	Payment Voucher	PV
11	Membership Payment	MP
12	Credit Note	CN

NEW TYPES TO ADD:

13	Cheque	CHQ
14	Deposit Slip	DS
15	Petty Cash Voucher	PCV
16	Journal Voucher	JV
17	Bank Transfer	BT
18	Debit Note	DN
19	Receipt Voucher	RV

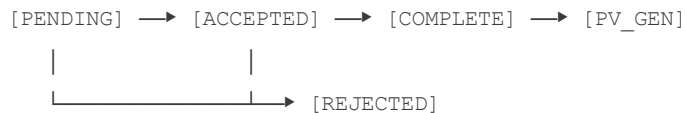
1.3.3 Voucher Status (Workflow)

voucher_status	
id	INT PK AUTO_INCREMENT
name	VARCHAR(50) NOT NULL
code	VARCHAR(20) UNIQUE
sequence	INT (for workflow order)
is_active	BOOLEAN DEFAULT TRUE

EXISTING DATA:

ID	Name	Code	Sequence
1	Pending	PENDING	1
2	Accepted	ACCEPTED	2
3	Rejected	REJECTED	99
4	Complete	COMPLETE	3
5	Payment Voucher Generated	PV_GEN	4

WORKFLOW:



1.3.4 Voucher Item (Recursive - Your Design)

voucher_item	
id	INT PK AUTO_INCREMENT
amount	DOUBLE
description	VARCHAR(500)
entered_date	DATETIME
voucher_id	INT FK → voucher.id NOT NULL
voucher_item_id	INT FK → voucher_item.id (SELF-REFERENCE)
is_active	TINYINT DEFAULT 1

INDEXES:

- PRIMARY KEY (id)
- INDEX idx_vi_voucher (voucher_id)

- INDEX idx_vi_parent (voucher_item_id)

NOTE: voucher_item_id is RECURSIVE - allows nested line items

1.3.5 Voucher Attachment

voucher_attachment	
id	INT PK AUTO_INCREMENT
file_name	TEXT NOT NULL
file_path	TEXT
file_type	VARCHAR(50)
file_size	BIGINT
voucher_id	INT FK → voucher.id NOT NULL
uploaded_at	DATETIME
uploaded_by	INT FK → user_login.id

1.3.6 Voucher Approval Manager

voucher_approval_manager	
id	INT PK AUTO_INCREMENT
voucher_id	INT FK → voucher.id NOT NULL
approval_level_id	INT FK → approval_level.id NOT NULL
approval_status_id	INT FK → approval_status.id NOT NULL
login_session_id	INT FK → login_session.id NOT NULL
approved_date	DATETIME
comment	TEXT

INDEXES:

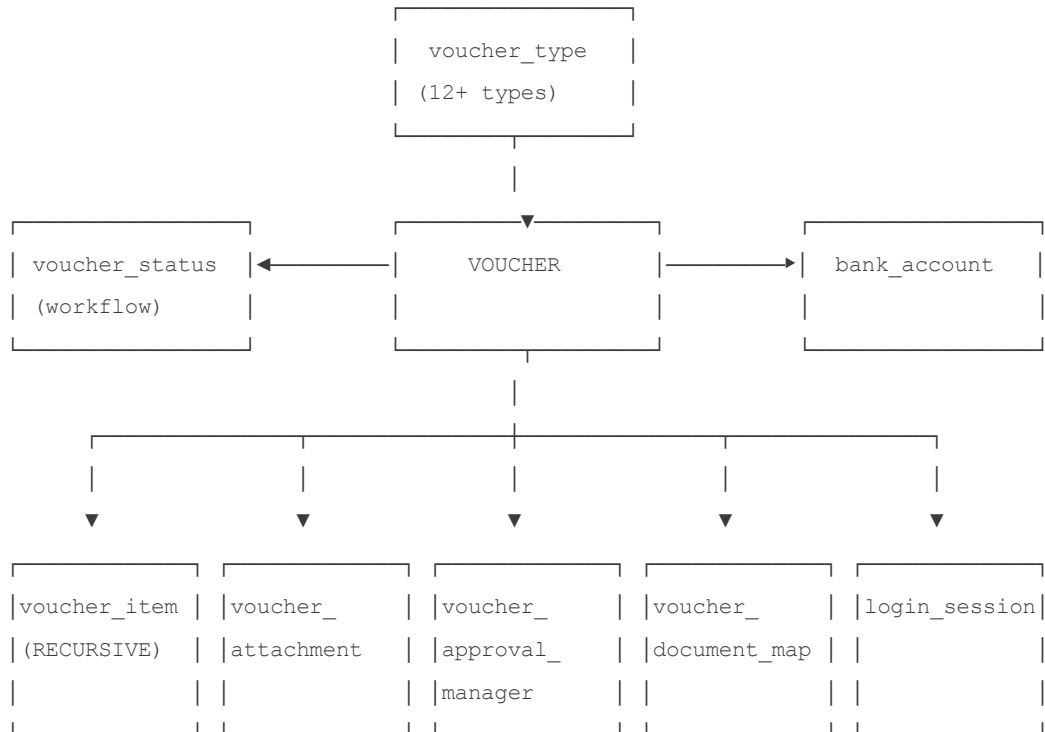
- PRIMARY KEY (id)
- INDEX idx_vam_voucher (voucher_id)
- INDEX idx_vam_status (approval_status_id)

1.3.7 Voucher Document Map

voucher_document_map	
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id	INT PK AUTO_INCREMENT
voucher_id	INT FK → voucher.id NOT NULL
document_id	INT FK → universal_user_document.id
created_at	DATETIME

Voucher Model ERD



SECTION 2: MASTER DATA TABLES

2.1 General User Profile

general_user_profile	
id	INT PK AUTO_INCREMENT
nic	VARCHAR(20) UNIQUE
first_name	VARCHAR(100)

last_name	VARCHAR(100)	
full_name	VARCHAR(255)	
date_of_birth	DATE	
gender_id	INT FK → gender.id	
email	VARCHAR(255)	
mobile	VARCHAR(20)	
address	TEXT	
city_id	INT FK → city.id	
district_id	INT FK → district.id	
province_id	INT FK → province.id	
country_id	INT FK → country.id	
education_level_id	INT FK → education_level.id	
profession_id	INT FK → profession.id	
profile_image	TEXT	
is_active	TINYINT DEFAULT 1	
created_at	DATETIME	
updated_at	DATETIME	

INDEXES:

- PRIMARY KEY (id)
- UNIQUE (nic)
- INDEX idx_gup_email (email)
- INDEX idx_gup_mobile (mobile)

2.2 General Organization Profile

general_organization_profile		
id	INT PK AUTO_INCREMENT	
registration_no	VARCHAR(50) UNIQUE	
org_name	VARCHAR(255) NOT NULL	
org_type_id	INT FK → organization_type.id	
email	VARCHAR(255)	
phone	VARCHAR(20)	
fax	VARCHAR(20)	
website	VARCHAR(255)	
address	TEXT	
city_id	INT FK → city.id	

district_id	INT FK → district.id
province_id	INT FK → province.id
country_id	INT FK → country.id
logo	TEXT
is_active	TINYINT DEFAULT 1
created_at	DATETIME
updated_at	DATETIME

INDEXES:

- PRIMARY KEY (id)
- UNIQUE (registration_no)
- INDEX idx_gop_name (org_name)

2.3 Bank Account

bank_account	
id	INT PK AUTO_INCREMENT
account_number	VARCHAR(50) UNIQUE NOT NULL
account_name	VARCHAR(255)
bank_id	INT FK → bank.id
branch_name	VARCHAR(255)
account_type_id	INT FK → account_type.id
currency_id	INT FK → currency.id
opening_balance	DECIMAL(18,2) DEFAULT 0.00
current_balance	DECIMAL(18,2) DEFAULT 0.00
org_profile_id	INT FK → general_organization_profile.id
is_active	TINYINT DEFAULT 1
created_at	DATETIME
updated_at	DATETIME

SECTION 3: SESSION & ACCESS CONTROL TABLES

3.1 User Login

user_login	
id	INT PK AUTO_INCREMENT
username	VARCHAR(255) UNIQUE NOT NULL
password	VARCHAR(255) NOT NULL (hashed)
is_active	TINYINT DEFAULT 1
max_login_attempt	INT DEFAULT 5
count_attempt	INT DEFAULT 0
user_role_id	INT FK → user_role.id
general_user_profile_id	INT FK → general_user_profile.id
general_organization_profile_id	INT FK → general_org_profile.id
last_login_at	DATETIME
password_changed_at	DATETIME
updated_at	DATETIME

INDEXES:

- PRIMARY KEY (id)
- UNIQUE (username)
- INDEX idx_ul_role (user_role_id)
- INDEX idx_ul_profile (general_user_profile_id)

3.2 Login Session

login_session	
id	INT PK AUTO_INCREMENT
user_login_id	INT FK → user_login.id
user_login_group_id	INT FK → user_login_group.id
general_organization_profile_id	INT FK → general_org_profile.id
start_time	DATETIME NOT NULL
end_time	DATETIME
ip	VARCHAR(50)
user_agent	VARCHAR(500)
is_active	BOOLEAN DEFAULT TRUE

INDEXES:

- PRIMARY KEY (id)
- INDEX idx_ls_user (user_login_id)
- INDEX idx_ls_start (start_time)

3.3 User Role

user_role	
id	INT PK AUTO_INCREMENT
name	VARCHAR(100) NOT NULL
code	VARCHAR(50) UNIQUE
description	TEXT
is_active	BOOLEAN DEFAULT TRUE

EXISTING DATA:

ID	Name	Code
1	Admin	ADMIN
2	Member	MEMBER
4	Accountant	ACCT
5	Finance Controller	FIN_CTRL
6	Chairman	CHAIRMAN
7	Supplier	SUPPLIER
8	Data Controller	DATA_CTRL
9	Department Head	DEPT_HEAD

3.4 User Login Group

user_login_group	
id	INT PK AUTO_INCREMENT
user_login_id	INT FK → user_login.id
user_role_id	INT FK → user_role.id
general_organization_profile_id	INT FK → general_org_profile.id
is_active	TINYINT DEFAULT 1

is_first_time	TINYINT DEFAULT 1
max_login_attempt	INT DEFAULT 5
count_attempt	INT DEFAULT 0

PURPOSE: Maps User → Role → Organization (multi-tenant support)

3.5 System Interface (GUI Registry)

system_interface	
id	INT PK AUTO_INCREMENT
interface_name	VARCHAR(100) NOT NULL
display_name	VARCHAR(100)
url	VARCHAR(255)
icon	VARCHAR(100)
interface_menu_id	INT FK → interface_menu.id
interface_sub_menu_id	INT FK → interface_sub_menu.id
sequence	INT (display order)
status	TINYINT DEFAULT 1

FINANCE INTERFACES TO ADD:

ID	Interface Name	Display Name	Menu
101	fin_dashboard	Finance Dashboard	Finance
102	fin_coa_list	Chart of Accounts	COA
103	fin_coa_create	Create Account	COA
104	fin_coa_edit	Edit Account	COA
105	fin_journal_list	Journal Entries	Journal
106	fin_journal_create	Create Journal	Journal
107	fin_journal_post	Post Journal	Journal
108	fin_voucher_list	Vouchers	Voucher
109	fin_voucher_create	Create Voucher	Voucher
110	fin_trial_balance	Trial Balance	Reports
111	fin_balance_sheet	Balance Sheet	Reports
112	fin_income_stmt	Income Statement	Reports
113	fin_cash_flow	Cash Flow Statement	Reports
114	fin_period_close	Close Period	Admin

115	fin_year_close	Close Year	Admin	
116	fin_bank_recon	Bank Reconciliation	Bank	

3.6 User Role Has System Interface

user_role_has_system_interface	
id	INT PK AUTO_INCREMENT
user_role_id	INT FK → user_role.id NOT NULL
system_interface_id	INT FK → system_interface.id NOT NULL
can_view	BOOLEAN DEFAULT TRUE
can_create	BOOLEAN DEFAULT FALSE
can_edit	BOOLEAN DEFAULT FALSE
can_delete	BOOLEAN DEFAULT FALSE
is_active	BOOLEAN DEFAULT TRUE

EXAMPLE MAPPINGS:

Role	Interface	View	Create	Edit	Delete
Accountant	fin_coa_list	✓	✓	✓	✗
Accountant	fin_journal_list	✓	✓	✓	✗
Accountant	fin_journal_post	✓	✗	✗	✗
Fin Ctrl	fin_journal_post	✓	✓	✓	✗
Fin Ctrl	fin_period_close	✓	✓	✗	✗
Admin	ALL	✓	✓	✓	✓

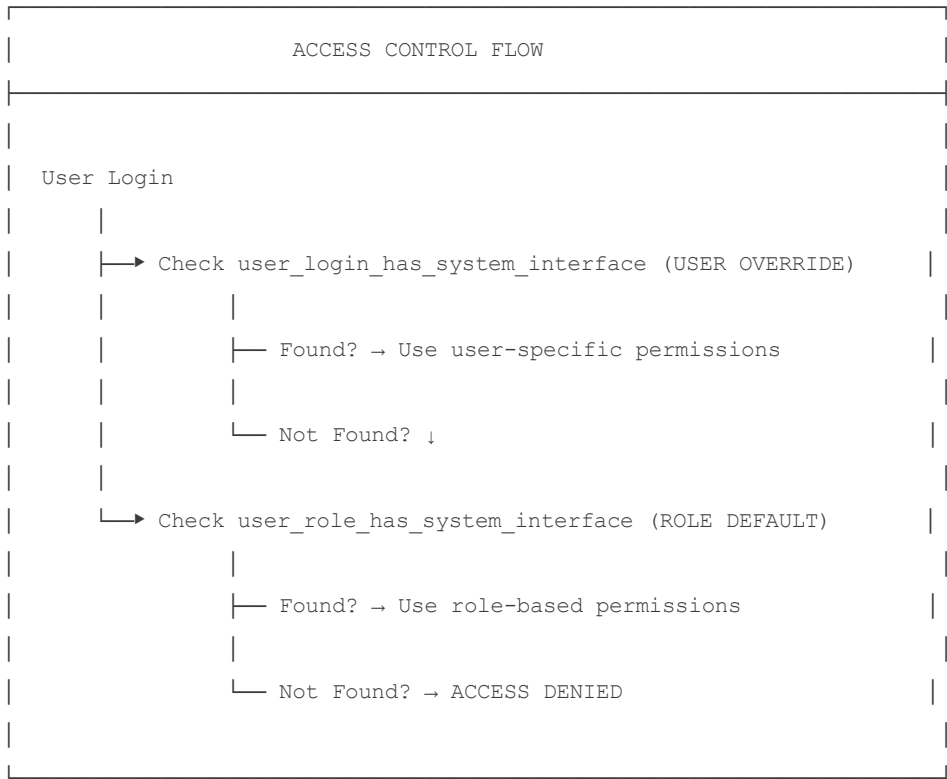
3.7 User Login Has System Interface

user_login_has_system_interface	
id	INT PK AUTO_INCREMENT
user_login_id	INT FK → user_login.id NOT NULL
system_interface_id	INT FK → system_interface.id NOT NULL
can_view	BOOLEAN DEFAULT TRUE
can_create	BOOLEAN DEFAULT FALSE
can_edit	BOOLEAN DEFAULT FALSE
can_delete	BOOLEAN DEFAULT FALSE

is_active	BOOLEAN DEFAULT TRUE
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PURPOSE: User-specific interface access (overrides role-based)

Access Control Flow



SECTION 4: LOGGING TABLES

4.1 Package Manager

package_manager	
id	INT PK AUTO_INCREMENT
name	VARCHAR(100) NOT NULL
code	VARCHAR(50) UNIQUE
description	TEXT

is_active	BOOLEAN DEFAULT TRUE
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EXISTING DATA:

ID	Name	Code
1	User Management	USER
2	Organization Management	ORG
3	Member Management	MEMBER
4	Voucher Management	VOUCHER
5	Spot Manager	SPOT
6	Location Management	LOCATION
7	Privillage Management	PRIV
8	Security Management	SECURITY
9	Loan_Management	LOAN

NEW TO ADD:

10	Finance Management	FINANCE
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4.2 Table Manager

table_manager	
id	INT PK AUTO_INCREMENT
name	VARCHAR(100) NOT NULL (table name)
display_name	VARCHAR(100)
package_manager_id	INT FK → package_manager.id NOT NULL
is_active	BOOLEAN DEFAULT TRUE

EXISTING DATA (Sample):

ID	Name	Package
1	general_user_profile	User Management
2	education_level	User Management
3	gop_has_member	User Management
4	gender	User Management
5	profeession	User Management
6	designation	User Management

FINANCE TABLES TO ADD:

ID	Name	Package
101	chart_of_account	Finance Management
102	fin_journal_header	Finance Management
103	fin_journal_line	Finance Management
104	fin_transaction_entry	Finance Management
105	fin_fiscal_year	Finance Management
106	fin_fiscal_period	Finance Management
...		

4.3 Data Changed Log Manager

data_changed_log_manager	
id	INT PK AUTO_INCREMENT
table_manager_id	INT FK → table_manager.id NOT NULL
user_login_id	INT FK → user_login.id NOT NULL
referance	VARCHAR(100) (record primary key)
attribute_name	VARCHAR(100) (field that changed)
old_value	TEXT
new_value	TEXT
action	ENUM('INSERT', 'UPDATE', 'DELETE')
comment	VARCHAR(255)
date	DATETIME NOT NULL

INDEXES:

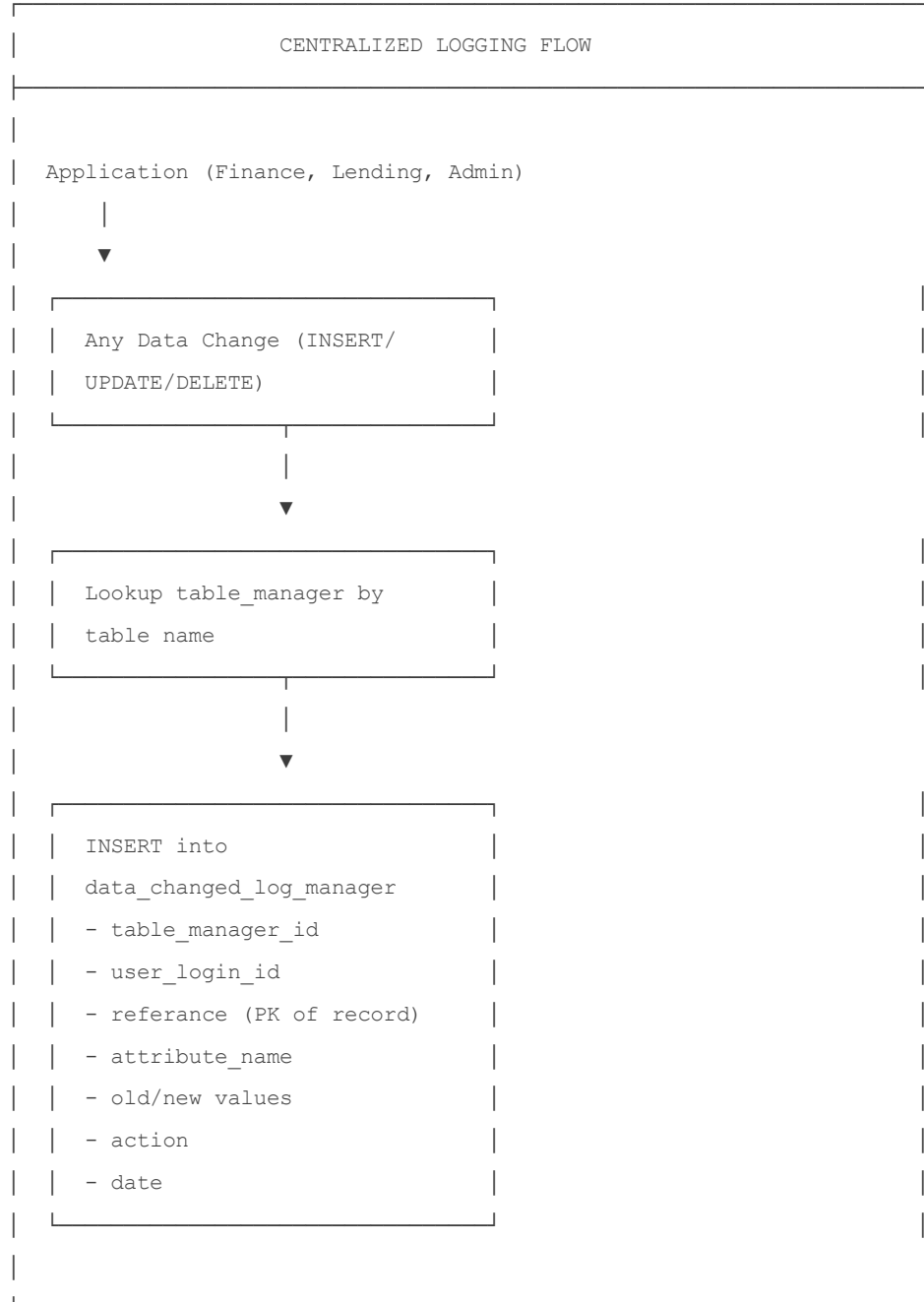
- PRIMARY KEY (id)
- INDEX idx_dcl_table (table_manager_id)
- INDEX idx_dcl_user (user_login_id)
- INDEX idx_dcl_date (date)
- INDEX idx_dcl_ref (referance)

EXAMPLE LOG ENTRIES:

table_manager	user	ref	attribute	old	new	action
chart_of_account	1	101	account_name	NULL	Petty Cash	INSERT

chart_of_account	1	101	account_name	Petty Cash	Office Cash	UPDATE
fin_journal_header	2	501	status	DRAFT	POSTED	UPDATE

Centralized Logging Flow



SECTION 5: NEW FINANCE TABLES (15 Tables)

5.1 fin_partner_type

fin_partner_type	
id	BIGINT PK AUTO_INCREMENT
type_code	VARCHAR(20) UNIQUE NOT NULL
type_name	VARCHAR(100) NOT NULL
is_active	BOOLEAN DEFAULT TRUE

SEED DATA:

ID	Code	Name
1	VENDOR	Vendor/Supplier
2	CUSTOMER	Customer
3	BOTH	Both

5.2 fin_trading_partner

fin_trading_partner	
id	BIGINT PK AUTO_INCREMENT
partner_code	VARCHAR(50) UNIQUE NOT NULL
partner_type_id	BIGINT FK → fin_partner_type.id NOT NULL
entity_type	ENUM('INDIVIDUAL', 'ORGANIZATION') NOT NULL
user_profile_id	INT FK → general_user_profile.id
org_profile_id	INT FK → general_organization_profile.id
default_coa_id	BIGINT FK → chart_of_account.id
credit_limit	DECIMAL(18,2) DEFAULT 0.00
payment_term_id	BIGINT FK → fin_payment_term.id
tax_id	VARCHAR(50)
is_active	BOOLEAN DEFAULT TRUE
created_at	DATETIME
updated_at	DATETIME

created_by	BIGINT
updated_by	BIGINT

CONSTRAINT:

- CHECK (user_profile_id IS NOT NULL OR org_profile_id IS NOT NULL)

5.3 fin_payment_term

fin_payment_term	
id	BIGINT PK AUTO_INCREMENT
term_code	VARCHAR(20) UNIQUE NOT NULL
term_name	VARCHAR(100) NOT NULL
days	INT NOT NULL DEFAULT 0
is_active	BOOLEAN DEFAULT TRUE

SEED DATA:

ID	Code	Name	Days
1	IMMEDIATE	Immediate Payment	0
2	NET7	Net 7 Days	7
3	NET15	Net 15 Days	15
4	NET30	Net 30 Days	30
5	NET60	Net 60 Days	60
6	NET90	Net 90 Days	90

5.4 fin_fiscal_year

fin_fiscal_year	
id	BIGINT PK AUTO_INCREMENT
year_code	VARCHAR(20) UNIQUE NOT NULL
year_name	VARCHAR(100) NOT NULL
start_date	DATE NOT NULL
end_date	DATE NOT NULL
is_closed	BOOLEAN DEFAULT FALSE

is_active	BOOLEAN DEFAULT TRUE
created_at	DATETIME
closed_at	DATETIME
closed_by	BIGINT FK → user_login.id

EXAMPLE:

ID	Code	Name	Start	End
1	FY2025-26	Fiscal 2025-26	2025-04-01	2026-03-31
2	FY2026-27	Fiscal 2026-27	2026-04-01	2027-03-31

5.5 fin_fiscal_period

fin_fiscal_period	
id	BIGINT PK AUTO_INCREMENT
fiscal_year_id	BIGINT FK → fin_fiscal_year.id NOT NULL
period_number	INT NOT NULL
period_name	VARCHAR(50) NOT NULL
start_date	DATE NOT NULL
end_date	DATE NOT NULL
status	ENUM('OPEN', 'CLOSED', 'LOCKED') DEFAULT 'OPEN'
is_adjustment	BOOLEAN DEFAULT FALSE
closed_at	DATETIME
closed_by	BIGINT FK → user_login.id

CONSTRAINT:

- UNIQUE (fiscal_year_id, period_number)

EXAMPLE (FY2025-26):

Period	Name	Start	End	Status
1	April 2025	2025-04-01	2025-04-30	CLOSED
2	May 2025	2025-05-01	2025-05-31	CLOSED
3	June 2025	2025-06-01	2025-06-30	OPEN
...	...	...	...	...
12	March 2026	2026-03-01	2026-03-31	OPEN

5.6 fin_currency

fin_currency	
id	BIGINT PK AUTO_INCREMENT
currency_code	VARCHAR(3) UNIQUE NOT NULL
currency_name	VARCHAR(100) NOT NULL
currency_symbol	VARCHAR(10)
decimal_places	INT DEFAULT 2
is_base	BOOLEAN DEFAULT FALSE
is_active	BOOLEAN DEFAULT TRUE

SEED DATA:

ID	Code	Name	Symbol	Decimals	Is Base
1	LKR	Sri Lankan Rupee	Rs.	2	TRUE
2	USD	US Dollar	\$	2	FALSE
3	EUR	Euro	€	2	FALSE
4	GBP	British Pound	£	2	FALSE

5.7 fin_exchange_rate

fin_exchange_rate	
id	BIGINT PK AUTO_INCREMENT
currency_id	BIGINT FK → fin_currency.id NOT NULL
effective_date	DATE NOT NULL
rate_to_base	DECIMAL(18,6) NOT NULL
source	VARCHAR(100)
created_at	DATETIME
created_by	BIGINT

CONSTRAINT:

- UNIQUE (currency_id, effective_date)

EXAMPLE:

Currency	Date	Rate to LKR
-----	-----	-----
USD	2026-02-01	320.50
USD	2026-02-02	321.00
EUR	2026-02-01	345.75

5.8 fin_journal_type

fin_journal_type	
id	BIGINT PK AUTO_INCREMENT
type_code	VARCHAR(20) UNIQUE NOT NULL
type_name	VARCHAR(100) NOT NULL
description	TEXT
is_active	BOOLEAN DEFAULT TRUE

SEED DATA:

ID	Code	Name
----	-----	-----
1	GJ	General Journal
2	SJ	Sales Journal
3	PJ	Purchase Journal
4	CRJ	Cash Receipts Journal
5	CPJ	Cash Payments Journal
6	BJ	Bank Journal
7	ADJ	Adjustment Journal

5.9 fin_journal_header

fin_journal_header	
id	BIGINT PK AUTO_INCREMENT
journal_number	VARCHAR(50) UNIQUE NOT NULL
journal_type_id	BIGINT FK → fin_journal_type.id NOT NULL
fiscal_period_id	BIGINT FK → fin_fiscal_period.id NOT NULL

journal_date	DATE NOT NULL	
reference	VARCHAR(100)	
narration	TEXT	
voucher_id	INT FK → voucher.id	
trading_partner_id	BIGINT FK → fin_trading_partner.id	
currency_id	BIGINT FK → fin_currency.id	
exchange_rate	DECIMAL(18,6) DEFAULT 1.000000	
total_debit	DECIMAL(18,2) NOT NULL DEFAULT 0.00	
total_credit	DECIMAL(18,2) NOT NULL DEFAULT 0.00	
status	ENUM('DRAFT','POSTED','REVERSED') DEFAULT 'DRAFT'	
posted_at	DATETIME	
posted_by	BIGINT FK → user_login.id	
reversed_by_id	BIGINT FK → fin_journal_header.id	
is_auto_generated	BOOLEAN DEFAULT FALSE	
created_at	DATETIME	
created_by	BIGINT FK → user_login.id	

CONSTRAINT:

- CHECK (total_debit = total_credit)

5.10 fin_journal_line

fin_journal_line		
id	BIGINT PK AUTO_INCREMENT	
journal_header_id	BIGINT FK → fin_journal_header.id NOT NULL	
line_number	INT NOT NULL	
account_id	BIGINT FK → chart_of_account.id NOT NULL	
credit_or_debit_id	INT FK → credit_or_debit.id NOT NULL	
amount	DECIMAL(18,2) NOT NULL	
base_amount	DECIMAL(18,2) NOT NULL	
description	VARCHAR(500)	
cost_center_id	BIGINT FK → fin_cost_center.id	
trading_partner_id	BIGINT FK → fin_trading_partner.id	

CONSTRAINT:

- UNIQUE (journal_header_id, line_number)

- CHECK (amount > 0)

5.11 fin_transaction_entry

fin_transaction_entry	
id	BIGINT PK AUTO_INCREMENT
voucher_id	INT FK → voucher.id
journal_header_id	BIGINT FK → fin_journal_header.id
account_id	BIGINT FK → chart_of_account.id NOT NULL
credit_or_debit_id	INT FK → credit_or_debit.id NOT NULL
amount	DECIMAL(18,2) NOT NULL
base_amount	DECIMAL(18,2) NOT NULL
currency_id	BIGINT FK → fin_currency.id
exchange_rate	DECIMAL(18,6) DEFAULT 1.000000
description	VARCHAR(500)
trading_partner_id	BIGINT FK → fin_trading_partner.id
cost_center_id	BIGINT FK → fin_cost_center.id
fiscal_period_id	BIGINT FK → fin_fiscal_period.id NOT NULL
transaction_date	DATE NOT NULL
is_posted	BOOLEAN DEFAULT FALSE
posted_at	DATETIME
posted_by	BIGINT FK → user_login.id
created_at	DATETIME
created_by	BIGINT

5.12 fin_voucher_type_account_map

fin_voucher_type_account_map	
id	BIGINT PK AUTO_INCREMENT
voucher_type_id	INT FK → voucher_type.id NOT NULL
account_id	BIGINT FK → chart_of_account.id NOT NULL
credit_or_debit_id	INT FK → credit_or_debit.id NOT NULL
is_default	BOOLEAN DEFAULT TRUE
description	VARCHAR(255)
sequence	INT DEFAULT 1

is_active	BOOLEAN DEFAULT TRUE
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EXAMPLE:

Voucher Type	Account	DR/CR	Description
Supplier Payment	Accounts Payable(2100)	DEBIT	Reduce liability
Supplier Payment	Bank Account (1112)	CREDIT	Cash outflow
Revenue Payment	Bank Account (1112)	DEBIT	Cash inflow
Revenue Payment	Revenue (4100)	CREDIT	Income

5.13 fin_cost_center

fin_cost_center	
id	BIGINT PK AUTO_INCREMENT
center_code	VARCHAR(50) UNIQUE NOT NULL
center_name	VARCHAR(255) NOT NULL
parent_id	BIGINT FK → fin_cost_center.id (RECURSIVE)
is_active	BOOLEAN DEFAULT TRUE
created_at	DATETIME
updated_at	DATETIME

EXAMPLE:

Code	Name	Parent
HQ	Head Office	NULL
HQ-F	Finance Dept	HQ
HQ-H	HR Dept	HQ
BR1	Colombo Branch	NULL
BR2	Kandy Branch	NULL

5.14 fin_bank_reconciliation

fin_bank_reconciliation	
id	BIGINT PK AUTO_INCREMENT

bank_account_id	INT FK → bank_account.id NOT NULL
fiscal_period_id	BIGINT FK → fin_fiscal_period.id NOT NULL
statement_date	DATE NOT NULL
statement_balance	DECIMAL(18,2) NOT NULL
book_balance	DECIMAL(18,2) NOT NULL
adjusted_balance	DECIMAL(18,2)
status	ENUM('IN_PROGRESS','COMPLETED') DEFAULT 'IN_PROGRESS'
reconciled_at	DATETIME
reconciled_by	BIGINT FK → user_login.id
notes	TEXT
created_at	DATETIME
created_by	BIGINT

5.15 fin_reconciliation_item

fin_reconciliation_item	
id	BIGINT PK AUTO_INCREMENT
reconciliation_id	BIGINT FK → fin_bank_reconciliation.id NOT NULL
journal_line_id	BIGINT FK → fin_journal_line.id
bank_statement_id	INT FK → bank_statement.id
status	ENUM('MATCHED','UNMATCHED','ADJUSTMENT')
matched_at	DATETIME
matched_by	BIGINT

SECTION 6: NEW COMMON TABLES (3 Tables)

6.1 com_application

com_application	
id	BIGINT PK AUTO_INCREMENT

app_code	VARCHAR(50) UNIQUE NOT NULL
app_name	VARCHAR(255) NOT NULL
description	TEXT
base_url	VARCHAR(255)
is_active	BOOLEAN DEFAULT TRUE
created_at	DATETIME

SEED DATA:

ID	Code	Name	URL
1	LENDING	Lending Application	https://lending.temcobank.com
2	CUSTOMER	Customer Portal	https://my.temcobank.com
3	ADMIN	Admin Panel	https://adminpanel.temcobank.com
4	FINANCE	Finance & Accounting	https://finance.temcobank.com

6.2 com_session_token

com_session_token	
id	BIGINT PK AUTO_INCREMENT
login_session_id	INT FK → login_session.id NOT NULL
application_id	BIGINT FK → com_application.id NOT NULL
access_token	VARCHAR(500) NOT NULL
refresh_token	VARCHAR(500)
token_type	VARCHAR(50) DEFAULT 'Bearer'
issued_at	DATETIME NOT NULL
expires_at	DATETIME NOT NULL
is_revoked	BOOLEAN DEFAULT FALSE
revoked_at	DATETIME
device_info	VARCHAR(500)
user_agent	VARCHAR(500)

INDEXES:

- INDEX idx_cst_session (login_session_id)
- INDEX idx_cst_token (access_token(255))
- INDEX idx_cst_expires (expires_at)

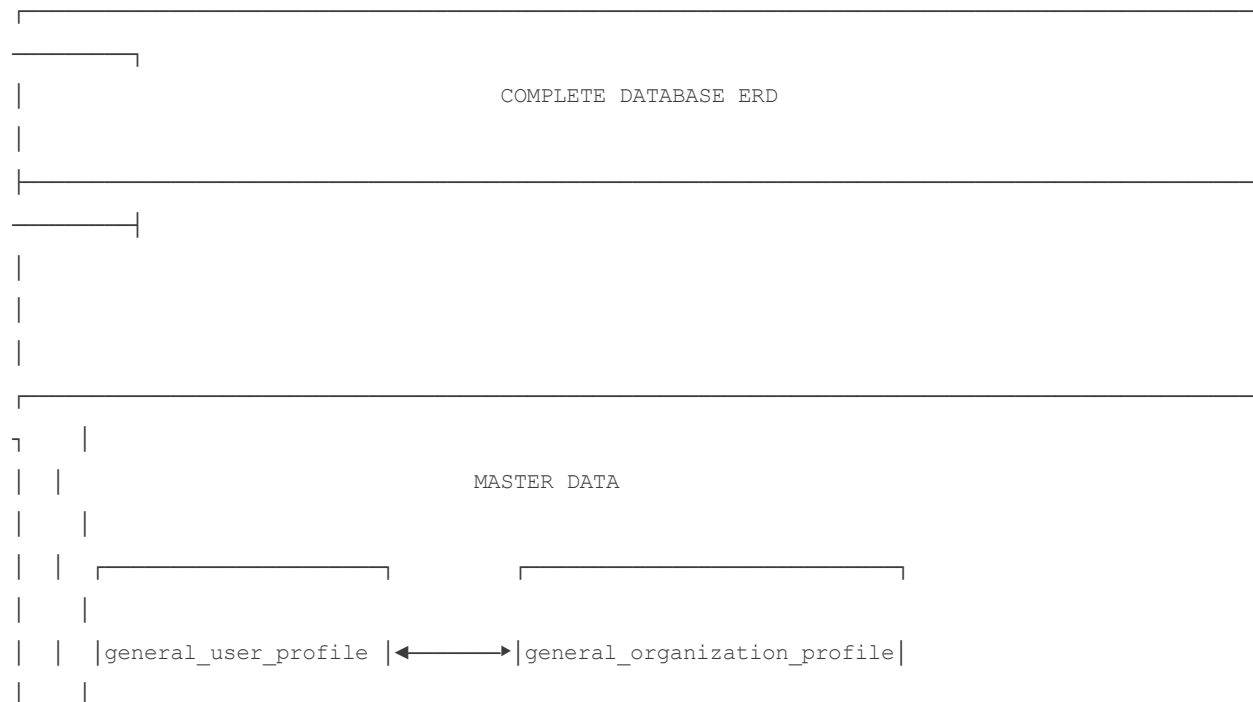
6.3 com_login_attempt

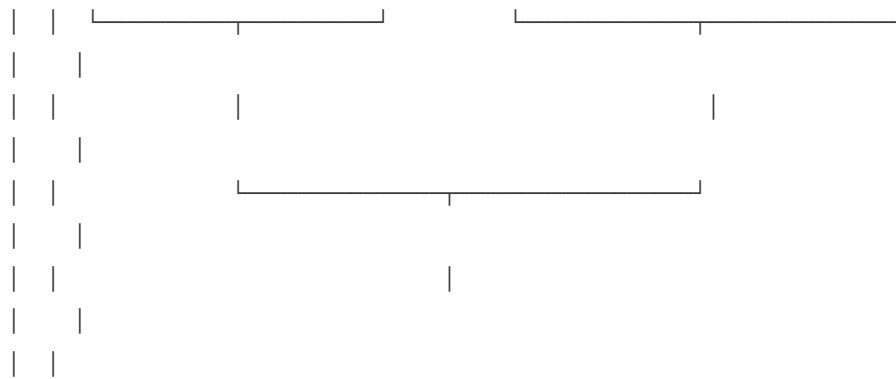
com_login_attempt	
id	BIGINT PK AUTO_INCREMENT
username	VARCHAR(255) NOT NULL
application_id	BIGINT FK → com_application.id
ip_address	VARCHAR(50)
user_agent	VARCHAR(500)
attempt_time	DATETIME NOT NULL
is_successful	BOOLEAN NOT NULL
failure_reason	VARCHAR(255)

INDEXES:

- INDEX idx_cla_username (username)
- INDEX idx_cla_time (attempt_time)
- INDEX idx_cla_ip (ip_address)

SECTION 7: COMPLETE ERD





The model's generation exceeded the maximum output token limit.

TEMCO Finance Database Architecture - Summary v2.0

COMPLETE TABLE LIST

YOUR CORE (11 Tables - Existing/Enhanced)

#	Table	Type	Key Feature
1	chart_of_account	Enhanced	RECURSIVE - parent_id self-reference
2	credit_or_debit	Existing	DEBIT/CREDIT lookup
3	voucher	Existing	Main document
4	voucher_type	Existing	Extensible types
5	voucher_status	Existing	Workflow states
6	voucher_item	Existing	RECURSIVE - nested items
7	voucher_attachment	Existing	File uploads
8	voucher_approval_manager	Existing	Multi-level approval

9	voucher_document_map	Existing	Document links
10	general_user_profile	Existing	Individual profiles
11	general_organization_profile	Existing	Org profiles

SESSION/ACCESS (7 Tables - Existing)

#	Table	Purpose
1	user_login	Credentials
2	login_session	Session tracking
3	user_role	Roles (Admin, Accountant, etc.)
4	user_login_group	User-Role-Org mapping
5	system_interface	GUI registry (every screen has ID)
6	user_role_has_system_interface	Role → Interface access
7	user_login_has_system_interface	User-specific overrides

LOGGING (3 Tables - Existing)

#	Table	Purpose
1	package_manager	Modules (add "Finance Management")
2	table_manager	Tables per module
3	data_changed_log_manager	Centralized audit log

NEW FINANCE (15 Tables)

#	Table	Purpose
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1	fin_partner_type	Vendor/Customer/Both
2	fin_trading_partner	Links to profiles
3	fin_payment_term	Net 30, etc.
4	fin_fiscal_year	Financial years
5	fin_fiscal_period	Months within year
6	fin_currency	Currency master
7	fin_exchange_rate	Daily rates
8	fin_journal_type	GJ, SJ, PJ, etc.
9	fin_journal_header	Journal entry header
10	fin_journal_line	Debit/credit lines
11	fin_transaction_entry	Voucher → COA value
12	fin_voucher_type_account_map	Default COA per voucher
13	fin_cost_center	RECURSIVE - cost tracking
14	fin_bank_reconciliation	Reconciliation header
15	fin_reconciliation_item	Matched items

NEW COMMON (3 Tables)

#	Table	Purpose
1	com_application	App registry
2	com_session_token	JWT tokens

3	com_login_attempt	Login audit
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KEY DESIGN PATTERNS

1. Recursive COA

chart_of_account.parent_id → chart_of_account.id
 5 ROOT accounts (Assets, Liabilities, Equity, Revenue, Expenses)
 UNLIMITED depth

2. Recursive Voucher Item

voucher_item.voucher_item_id → voucher_item.id
 Nested line items

3. Unified Document Model

ONE voucher table + voucher_type = ALL documents
 (Cheque, Payment, Receipt, Invoice, etc.)

4. Master Data Reuse

fin_trading_partner → general_user_profile (Individual)
 fin_trading_partner → general_organization_profile (Org)

5. Interface-Based Access

system_interface (GUI IDs)
 user_role_has_system_interface (Role defaults)
 user_login_has_system_interface (User overrides)

6. Centralized Logging

package_manager → table_manager → data_changed_log_manager

TRANSACTION FLOW

VOUCHER → fin_transaction_entry → chart_of_account
 ↓

credit_or_debit (DR/CR)

↓

fin_journal_header → fin_journal_line

TOTAL TABLES

Category	Count
Your Core	11
Session/Access	7
Logging	3
New Finance	15
New Common	3
TOTAL NEW	18

Ready for your review. Any changes needed before I create the SQL schema files?

Feedback submitted